Adaptive Gait Generation and Postural Stabilization Framework for Quadruped Robots via IMU Feedback Integration

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***Abstract*—This paper presents a robust, computationally efficient motion control framework engineered for articulated quadruped robots. Operating without exteroceptive vision systems, the framework achieves stable blind ambulation by leveraging proprioceptive feedback derived exclusively from an Inertial Measurement Unit (IMU). A strictly decoupled, dual-pipeline software architecture isolates the rhythmic pacing of a central pattern generator from the reactionary calculus of a proportional-integral-derivative (PID) stabilization node. By defining absolute spatial limits via inverse kinematics and shaping trajectories through piecewise quintic polynomials, the framework guarantees a solid morphological baseline. Crucially, dynamic disturbances are aggressively attenuated by applying phase-aware rotational compensations—$R_y(\theta)$ and $R_x(\phi)$—exclusively to the stance limbs. This highly selective logic forces the quadruped to counter-exert against gravitational and terrain-based anomalies, actively sustaining the Zero Moment Point (ZMP) within its support polygon. Preliminary tests conducted within the Gazebo simulation environment explicitly validate the system's efficacy in rejecting intense lateral disturbances while maintaining reliable home postures.**

I. INTRODUCTION

Legged robots possess the innate structural capacity to navigate discontinuous terrain and extreme topographies that inherently challenge conventional wheeled platforms. However, developing resilient locomotion policies capable of stabilizing a heavily articulated, multi-degree-of-freedom system against unmodeled disturbances remains a rigorous control challenge.

Contemporary solutions rely heavily on computationally intensive algorithms, such as Model Predictive Control (MPC) and Reinforcement Learning (RL), to estimate optimal zero-moment dynamics [1], [2]. While highly performant, these stochastic solvers predominantly dictate the use of specialized hardware accelerators or high-frequency processors. This computational overhead severely precludes deployment upon the low-power embedded edge devices ubiquitous in modern mechatronics.

Conversely, heuristic-driven Central Pattern Generators (CPGs) paired with reactive reflex feedback loops can successfully approximate biological stability at a fraction of the computational cost. When high-fidelity sensory telemetry—such as instantaneous base orientation measured by an Inertial Measurement Unit (IMU)—is directly integrated into real-time kinematic calculations, robust blind ambulation becomes trivialized.

This work proposes a refined, lightweight software architecture explicitly tailored for constrained robotic processing environments. By prioritizing a strict programmatic decoupling between rhythmic Cartesian trajectory generation and rapid-response inertial base stabilization, systemic cohesion is optimized. The framework deliberately isolates all PID formulations within a discrete mathematical node, proving that minimalist reactive control loops can effectively synthesize stable, multi-link continuous gaits.

II. System Architecture

The framework operates as an inherently distributed processing cluster managed entirely by Robot Operating System 2 (ROS 2). The core locomotion logic is brokered across two distinct process endpoints: a primary *Node Leg Controller* and an asynchronous *Node Balance Controller*. Communication between these modules is strictly limited to high-frequency diagnostic and corrective topics. This segregation guarantees hardware-level isolation, ensuring that rhythmic trajectory generation anomalies cannot preempt or block critical sensory compensations.

Fig. 1. Proposed System Architecture graph displaying the strict separation between the Gait Generation pipeline and the isolated PID stabilization loop within the ROS 2 environment.

A. State Representation

Global governance overlying the primary ROS 2 architecture is dictated by boolean gating, handling two discrete operating states:

- **HOME:** Defined exclusively as a static configuration lacking cyclic translation vectors. The controller computes uniform joint-space adjustments rigidly mapped to the resting stance. Base angular deviations translate proportionally into antagonistic torque compensations explicitly aimed at continuous postural rectification.

- **GAIT:** The dynamic state active during commanded planar translations. The platform executes cyclic polynomial trajectories, selectively applying spatial Cartesian corrections derived from the IMU exclusively to distal limbs classified within the *stance* phase. Swing legs inherently maintain zero-compensation matrices to ensure uninhibited stride clearance.

III. Kinematics and Gait Planning

A. Forward and Inverse Kinematics

To accurately mandate the position of a limb's endpoint (x, y, z) relative to the geometric origin of the base frame, a non-linear inverse kinematics (IK) model derives the necessary articular angles $(\theta_1, \theta_2, \theta_3)$ for the hip, thigh, and calf joints. The physical topology structuring the Cartesian workspace includes the platform length L, platform width W, and individual linkage spans l_1, l_2, and l_3.

The generalized mathematical solution avoids geometric singularities by enforcing strictly restricted bounding limits calculated dynamically during system boot initialization.

Fig. 2. The Denavit-Hartenberg (DH) geometric reference frame assigned to the three-degree-of-freedom quadruped limb.

B. Quintic Trajectory Formulation

It is mathematically imperative that physical ground impact velocities approach zero; violent mechanical shocks corrupt internal IMU localization loops and induce structural fatigue. Therefore, smooth, jerk-minimized foot trajectories are formulated by employing overlapping piecewise quintic polynomials rather than generic linear extrapolations.

For a specific foot transition evolving from resting coordinate A (lift-off) and traversing to coordinate D (touch-down) across constrained spanning time T_{swing}, an independent trajectory locus $p(t)$ enforces continuous physical boundaries in positional space, derivative velocity, and second-derivative acceleration.

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \tag{1}$$

By perfectly aligning adjacent spline limits exactly at $t = 0$ and $t = T_{swing}$, the iterative loops entirely preclude unmodeled physical resonance. Overlapping standard trotting patterns are generated efficiently via phase-shifting scalar arrays (e.g., 0.50 shifting applied to diagonal limb pairs).

IV. IMU-Based Postural Stabilization

Inertial corrections are computed asynchronously within the strictly isolated *Node Balance Controller*. The native hardware orientation vectors, specifically the localized Earth-referenced Pitch (θ) and Roll (ϕ), are ingested at high frequency.

Fig. 3. Control flow diagram visualizing the transformation pipeline processing raw IMU telemetry into discrete 100 Hz corrective polynomials.

A. Static Home Posture Correction

During the static phase, all limbs maintain sustained ground contact. High-frequency electrical noise from the IMU inherently poses a severe risk of jitter-induced mechanical resonance. To mitigate micro-oscillations, raw angular error telemetry cascades through an aggressive mathematical deadzone limits block:

$$e_{dz} = \begin{cases} e - dz & e > dz \\ 0 & |e| \leq dz \\ e + dz & e < -dz \end{cases} \tag{2}$$

The verified deviations are subsequently amplified by proportional tuning constants K_P. Because immediate, unscaled application could induce destructive torque spikes or singularity collapse across the Inverse Kinematics frame, the vectors are saturated algorithmically by absolute maximum positional caps. Finally, they are recursively smoothed via a first-order Low-Pass Filter (LPF), defined by a tunable filtering constant α:

$$H_k = \alpha H_{k-1} + (1 - \alpha) e_{scaled} \tag{3}$$

B. Phase-Aware Dynamic Gait Correction

While the robot is actively ambulating, naive rigid correction logic violently opposes intended forward locomotive kinematics. Consequently, sensory correction must be applied non-uniformly, in direct relation to the instantaneous stance status provided by the trajectory node.

The positional vector adjustment dictates a tri-dimensional rotational matrix $R = R_y(\theta) R_x(\phi)$. This equation is permitted to resolve exclusively when the limb altitude physically traverses below a predefined threshold boundary Z_{thresh}.

$$P_{adj} = R \cdot P_{raw} \tag{4}$$

Thus, if $|Z_{raw} - Z_{stance}| \leq Z_{thresh}$, the computational architecture conclusively confirms the mechanical foot is bearing gravitational load. The inverse kinematics engine immediately recalculates and actuates the adjusted internal angles, physically pushing against the anomalous terrain slope to right the base parallel to gravity.

V. Results and Diagnostic Data

A robust diagnostic data-publishing pipeline continuously normalizes and broadcasts internal software states (specifically across the `/diag/home/apply/*` topic topology) to objectively visualize algorithmic performance over active hardware tests.

A. Locomotion Viability in Simulation

To concretely prove algorithmic fidelity, physical hardware execution is preceded by exhaustive validation inside the rigid-body physics simulator, *Gazebo*. Dynamic lateral impact forces are intentionally applied against the mechanical chassis to evaluate disturbance rejection.

Fig. 4. Visual capture within Gazebo displaying the quadruped actively recovering its stable Zero Moment position following an unscheduled lateral disturbance.

The simulated environment verifies that the decoupled software architecture successfully avoids computational locking. Furthermore, the P-gain scaling mathematically guarantees sufficient rotational torque to reliably re-level the chassis.

B. Internal Control Graph Observability

Quantitative metric gathering confirms system efficiency. The implementation of standard deadzone gating successfully rejected continuous trivial analog IMU measurement noise, drastically reducing servo jitter and preserving sustained battery longevity.

VI. Conclusion and Future Work

The decoupled PID-balance and Central Pattern Generator formulation detailed within this framework exhibits a measurable improvement in guaranteeing physical stability over complex, unmapped terrain domains. By explicitly reserving cyclic step phase-shifts and quintic trajectory management to an isolated primary gait generator, the asynchronous secondary balance algorithm undeniably maintains unimpeded, real-time corrective authority over all active stance limbs.

Fig. 5. Empirical data plotted natively via ROS 2 PlotJuggler, mapping the mathematical reduction in high-frequency micro-oscillations derived from the LPF parameter tuning.

Immediate future programmatic iterations will deeply analyze the incorporation of Ground-Reaction Force (GRF) estimations, calculated via internal current sensors mapping absolute torque metrics against individual footfalls, enabling advanced pseudo-haptic terrain prediction topologies.

References

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